Venkata Lakshmi Parimala Pasupuleti

Lead AI Engineer · LangGraph · Multi-Agent · RAG · LLMOps · PropTech · Production AI · Python · TypeScript · AWS · Technical Roadmap · Mentorship

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PROFESSIONAL SUMMARY

Lead AI engineer with 8 years building production SaaS for property managers and owners — the exact domain RealPage serves. At Entrata (a direct RealPage competitor): **defined technical direction and engineering standards** for 20+ production modules, led three platform-level initiatives (Docker, AWS, production AI pipeline on **1.6M real financial records**), and mentored 5+ engineers. Built **AgentProbe**: 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix), 5-metric LLM-as-judge eval engine, CI/CD regression gates blocking production, and React observability dashboard. Built **Leasel**: agentic RAG over 1,240 legal PDFs, 93.6% precision, deterministic rules engine for fully auditable outputs — directly applicable to RealPage's lease AI. PropTech domain knowledge: lease processing, rent pricing workflows, owner reporting, maintenance automation — from 8 years of production incidents, not job descriptions. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 – Aug 2025

Research Associate — ML Pipelines · Anomaly Detection · Statistical Modeling

- Built end-to-end ML pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; automated statistical reporting with Python + Quarto driving **40% improvement** in stakeholder comprehension.

Entrata
Jan 2021 – Aug 2024

Senior Software Engineer — AI Technical Leadership · Agentic Systems · PropTech Domain · AWS · Mentorship

- **Defined technical direction and set engineering standards** for 20+ production modules end-to-end (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints — architecture, design patterns, implementation, rollout, adoption — **1M+ records, 100K+ MAU, 99.5% uptime**.
- Led **three platform-level initiatives**: Docker microservices rollout (50% reliability gain), AWS S3/RDS/Glue migration (100% PHI compliance, zero incidents), and Entrata's **first production AI pipeline on 1.6M real financial records** (PyTorch, OpenAI API) with quality gates, output validation, and HITL review before customer exposure.
- Mentored **5+ engineers**, conducted **300+ code reviews**; built HUD Section 8/HCV and LIHTC (Tax Credit) compliance modules with deterministic rules engine — eliminated **80% of manual workload** with full audit trails, zero compliance violations across 100K+ units.

Entrata
Jan 2019 – Dec 2020

Software Engineer — Compliance Automation · API Integrations · Data Pipelines

- Built compliance automation processing **millions of records** — income limits, household composition, asset verification thresholds in deterministic systems with full audit trails; built **API integrations and webhooks** across property management systems; 300+ peer code reviews; mentored 5+ junior engineers.

Entrata
Jan 2017 – Dec 2018

Associate Engineer — Full-Stack · Python · PostgreSQL · REST APIs

- Built and shipped full-stack features and SQL data layers (PHP, JavaScript, PostgreSQL) for property management workflows; 0→1 prototyping through production rollout across 100K+ unit portfolios.

PROJECTS

AgentProbe: Multi-Agent AI Eval Framework
MULTI-AGENT LLM-AS-JUDGE

Python · LangGraph · FastAPI · ChromaDB · OpenAI · Anthropic · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling and inter-agent state handoffs; **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with HITL feedback loops; **CI/CD regression gates blocking failing runs before production**; 15-endpoint FastAPI backend; React observability dashboard.

Leasel: Agentic Compliance Pipeline for Legal Documents
PROPTECH AGENTIC RAG

Python · LangGraph · ChromaDB · Deterministic Rules Engine · LLM-as-Judge · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision, 87.9% F1** on 1,240 complex legal PDFs — production RAG over domain documents: extraction, chunking, embedding, retrieval, deterministic rules engine for fully auditable high-stakes outputs; pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with LLM-as-judge and CI/CD regression gates — directly applicable to RealPage's lease document AI.

AI on FHIR: HIPAA-Compliant Medical Records NLP Platform
HEALTHCARE

Python · Flask · React · FHIR APIs · BERT · LLMs · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in clinical retrieval time — NLP platform translating natural-language queries (EN/ES) into FHIR API calls via LLMs + BERT; HIPAA-compliant RBAC, end-to-end from design to deployment.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

ML systems, statistical modeling, data engineering, AI pipelines, system design, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Languages & Backend: Python, JavaScript, TypeScript, SQL, PL/SQL, PHP, R; FastAPI, Django, Flask, REST APIs, webhooks, API integrations, RabbitMQ, event-driven distributed systems

AI & LLMOps: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, RAG Pipelines, Tool Calling, LLM-as-Judge, LLM Evaluation, AI observability, HITL Feedback Loops, Vector Stores (ChromaDB, pgvector), OpenAI API, Anthropic API, LangSmith (familiar), OpenAI Evals (familiar), PyTorch, BERT, NLP, prompt engineering, responsible AI

Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), GCP (familiar), Docker, Kubernetes, Microservices, CI/CD, GitHub Actions, PostgreSQL, MySQL, HIPAA, PHI compliance, technical roadmap definition, scalable architecture, design patterns, Agile/Scrum, Git, Linux